FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA

MINISTRY OF EDUCATION

GRADE 9 CHEMISTRY STUDENT TEXTBOOK

Grade 9 - Chemistry Official Student Textbook

SYNCHRONIZED CURRICULUM UNITS:

- Unit 1: Chemistry and Its Importance [Textbook Pages 1 - 30]
- Unit 2: Measurements and Scientific Calculations [Textbook Pages 31 - 64]
- Unit 3: Structure of the Atom [Textbook Pages 65 - 104]
- Unit 4: Periodic Classification of the Elements [Textbook Pages 105 - 142]
- Unit 5: Chemical Bonding and Intermolecular Forces [Textbook Pages 143 - 181]

GRADE 9 CHEMISTRY STUDENT TEXTBOOK - TABLE OF CONTENTS

UNIT 1: CHEMISTRY AND ITS IMPORTANCE [Textbook Pages 1 - 30]

1.1 Definition and Scope of Chemistry ..... Page 2
1.2 Chemistry in Industry, Medicine and Agriculture ..... Page 9
1.3 The Scientific Method in Chemistry ..... Page 16
1.4 Laboratory Safety, Hazard Symbols and Common Apparatus ..... Page 23

UNIT 2: MEASUREMENTS AND SCIENTIFIC CALCULATIONS [Textbook Pages 31 - 64]

2.1 Physical Quantities and the SI System of Units ..... Page 32
2.2 Scientific Notation, Metric Prefixes and Precision ..... Page 40
2.3 Significant Figures and Uncertainty in Measurement ..... Page 48
2.4 Dimensional Analysis [Factor-Label Method] ..... Page 56

UNIT 3: STRUCTURE OF THE ATOM [Textbook Pages 65 - 104]

3.1 Historical Development of the Atomic Model ..... Page 66
3.2 Subatomic Particles: Protons, Neutrons and Electrons ..... Page 75
3.3 Atomic Number, Mass Number, and Isotopes ..... Page 83
3.4 Bohr Model and Electron Energy Levels ..... Page 91
3.5 Valence Electrons and Electronic Stability ..... Page 98

UNIT 4: PERIODIC CLASSIFICATION OF THE ELEMENTS [Textbook Pages 105 - 142]

4.1 Historical Development and the Modern Periodic Law ..... Page 106
4.2 Periods, Groups, and Blocks of the Periodic Table ..... Page 114
4.3 Periodic Trends: Atomic Radius and Ionic Size ..... Page 123
4.4 Ionization Energy, Electron Affinity, and Electronegativity ..... Page 132

UNIT 5: CHEMICAL BONDING AND INTERMOLECULAR FORCES [Textbook Pages 143 - 181]

5.1 The Octet Rule and Why Atoms Combine ..... Page 144
5.2 Ionic [Electrovalent] Bonding and Properties ..... Page 151
5.3 Covalent Bonding and Molecular Geometry ..... Page 159
5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces ..... Page 168

1.1 Definition and Scope of Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 2 - Current Page: 1

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Curriculum Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.1 Definition and Scope of Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 2 - Current Page: 2

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Curriculum Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.1 Definition and Scope of Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 2 - Current Page: 3

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Curriculum Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.1 Definition and Scope of Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 2 - Current Page: 4

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Curriculum Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.1 Definition and Scope of Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 2 - Current Page: 5

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Curriculum Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.1 Definition and Scope of Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 2 - Current Page: 6

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Curriculum Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.1 Definition and Scope of Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 2 - Current Page: 7

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Curriculum Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.1 Definition and Scope of Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 2 - Current Page: 8

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Curriculum Focus: Chemistry as study of composition, structure, properties, and changes of matter; central role in physical and life sciences.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.2 Chemistry in Industry, Medicine and Agriculture

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 9 - Current Page: 9

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Curriculum Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.2 Chemistry in Industry, Medicine and Agriculture

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 9 - Current Page: 10

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Curriculum Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.2 Chemistry in Industry, Medicine and Agriculture

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 9 - Current Page: 11

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Curriculum Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.2 Chemistry in Industry, Medicine and Agriculture

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 9 - Current Page: 12

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Curriculum Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.2 Chemistry in Industry, Medicine and Agriculture

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 9 - Current Page: 13

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Curriculum Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.2 Chemistry in Industry, Medicine and Agriculture

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 9 - Current Page: 14

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Curriculum Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.2 Chemistry in Industry, Medicine and Agriculture

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 9 - Current Page: 15

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Curriculum Focus: Synthetic fertilizers, pharmaceuticals, plastics, soaps, metallurgy, and food preservation in Ethiopia.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.3 The Scientific Method in Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 16 - Current Page: 16

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Curriculum Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.3 The Scientific Method in Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 16 - Current Page: 17

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Curriculum Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.3 The Scientific Method in Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 16 - Current Page: 18

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Curriculum Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.3 The Scientific Method in Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 16 - Current Page: 19

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Curriculum Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.3 The Scientific Method in Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 16 - Current Page: 20

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Curriculum Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.3 The Scientific Method in Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 16 - Current Page: 21

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Curriculum Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.3 The Scientific Method in Chemistry

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 16 - Current Page: 22

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Curriculum Focus: Systematic observation, hypothesis formulation, controlled experimentation, law vs theory formulation.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.4 Laboratory Safety, Hazard Symbols and Common Apparatus

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 23 - Current Page: 23

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Curriculum Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.4 Laboratory Safety, Hazard Symbols and Common Apparatus

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 23 - Current Page: 24

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Curriculum Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.4 Laboratory Safety, Hazard Symbols and Common Apparatus

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 23 - Current Page: 25

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Curriculum Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.4 Laboratory Safety, Hazard Symbols and Common Apparatus

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 23 - Current Page: 26

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Curriculum Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.4 Laboratory Safety, Hazard Symbols and Common Apparatus

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 23 - Current Page: 27

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Curriculum Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.4 Laboratory Safety, Hazard Symbols and Common Apparatus

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 23 - Current Page: 28

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Curriculum Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.4 Laboratory Safety, Hazard Symbols and Common Apparatus

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 23 - Current Page: 29

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Curriculum Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

1.4 Laboratory Safety, Hazard Symbols and Common Apparatus

Unit 1 [Pages 1-30] - Topic Starts on Textbook Page 23 - Current Page: 30

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Curriculum Focus: Beakers, flasks, graduated cylinders, Bunsen burners, pipettes, and safe handling of corrosive and toxic chemicals.

- Definition & Scope of Chemistry: Chemistry is the central physical science studying the composition, structure, properties, and transformation of matter. Core branches: Organic, Inorganic, Physical, Analytical, and Biochemistry.

- Chemistry in Industry, Medicine & Agriculture: Synthetic fertilizers [urea, DAP] and pesticides boost crop yield. Pharmaceuticals and vaccines combat tropical diseases. Metallurgy, polymers, soaps, and cement industries drive economic infrastructure.

- Scientific Method in Chemistry: Observation Posing testable research question Formulating hypothesis Controlled experimentation [variables] Data collection & analysis Formulation of Scientific Law [summarizes observations] vs Scientific Theory [explains underlying mechanisms].

- Laboratory Safety & Common Apparatus: Safety goggles, lab coats, fume hoods. Apparatus: Beakers, Erlenmeyer flasks, graduated cylinders [volume measurement], burettes & pipettes [precise titration], Bunsen burners. Hazard Symbols: Flammable [flame], Toxic [skull & crossbones], Corrosive [corroding surface], Explosive [exploding bomb].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemistry explores atomic and molecular transformations. Applying the scientific method enables industrial chemical manufacturing, pharmaceutical design, and agricultural soil management.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Chemistry connects physics with biological sciences ["the central science"].
- A scientific law describes WHAT happens; a scientific theory explains WHY it happens.
- Volumetric measurements require reading the bottom of the liquid meniscus at eye level.

Ethiopian Context: Mugher and Derba cement factories, Wonji sugar refinery, and national fertilizer distribution depots in Ethiopia apply industrial chemistry principles.

2.1 Physical Quantities and the SI System of Units

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 32 - Current Page: 31

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Curriculum Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.1 Physical Quantities and the SI System of Units

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 32 - Current Page: 32

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Curriculum Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.1 Physical Quantities and the SI System of Units

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 32 - Current Page: 33

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Curriculum Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.1 Physical Quantities and the SI System of Units

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 32 - Current Page: 34

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Curriculum Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.1 Physical Quantities and the SI System of Units

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 32 - Current Page: 35

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Curriculum Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.1 Physical Quantities and the SI System of Units

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 32 - Current Page: 36

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Curriculum Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.1 Physical Quantities and the SI System of Units

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 32 - Current Page: 37

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Curriculum Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.1 Physical Quantities and the SI System of Units

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 32 - Current Page: 38

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Curriculum Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.1 Physical Quantities and the SI System of Units

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 32 - Current Page: 39

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Curriculum Focus: Base SI units [meter, kilogram, second, kelvin, mole] and derived units [volume, density, concentration].

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.2 Scientific Notation, Metric Prefixes and Precision

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 40 - Current Page: 40

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Curriculum Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.2 Scientific Notation, Metric Prefixes and Precision

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 40 - Current Page: 41

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Curriculum Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.2 Scientific Notation, Metric Prefixes and Precision

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 40 - Current Page: 42

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Curriculum Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.2 Scientific Notation, Metric Prefixes and Precision

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 40 - Current Page: 43

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Curriculum Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.2 Scientific Notation, Metric Prefixes and Precision

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 40 - Current Page: 44

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Curriculum Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.2 Scientific Notation, Metric Prefixes and Precision

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 40 - Current Page: 45

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Curriculum Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.2 Scientific Notation, Metric Prefixes and Precision

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 40 - Current Page: 46

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Curriculum Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.2 Scientific Notation, Metric Prefixes and Precision

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 40 - Current Page: 47

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Curriculum Focus: Expressing large/small numbers in exponential form [N x 10^n], mega, kilo, centi, milli, micro, nano prefixes.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.3 Significant Figures and Uncertainty in Measurement

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 48 - Current Page: 48

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Curriculum Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.3 Significant Figures and Uncertainty in Measurement

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 48 - Current Page: 49

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Curriculum Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.3 Significant Figures and Uncertainty in Measurement

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 48 - Current Page: 50

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Curriculum Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.3 Significant Figures and Uncertainty in Measurement

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 48 - Current Page: 51

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Curriculum Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.3 Significant Figures and Uncertainty in Measurement

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 48 - Current Page: 52

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Curriculum Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.3 Significant Figures and Uncertainty in Measurement

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 48 - Current Page: 53

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Curriculum Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.3 Significant Figures and Uncertainty in Measurement

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 48 - Current Page: 54

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Curriculum Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.3 Significant Figures and Uncertainty in Measurement

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 48 - Current Page: 55

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Curriculum Focus: Rules for counting significant figures, rounding off rules in addition, subtraction, multiplication, and division.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.4 Dimensional Analysis [Factor-Label Method]

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 56 - Current Page: 56

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Curriculum Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.4 Dimensional Analysis [Factor-Label Method]

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 56 - Current Page: 57

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Curriculum Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.4 Dimensional Analysis [Factor-Label Method]

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 56 - Current Page: 58

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Curriculum Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.4 Dimensional Analysis [Factor-Label Method]

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 56 - Current Page: 59

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Curriculum Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.4 Dimensional Analysis [Factor-Label Method]

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 56 - Current Page: 60

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Curriculum Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.4 Dimensional Analysis [Factor-Label Method]

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 56 - Current Page: 61

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Curriculum Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.4 Dimensional Analysis [Factor-Label Method]

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 56 - Current Page: 62

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Curriculum Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.4 Dimensional Analysis [Factor-Label Method]

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 56 - Current Page: 63

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Curriculum Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

2.4 Dimensional Analysis [Factor-Label Method]

Unit 2 [Pages 31-64] - Topic Starts on Textbook Page 56 - Current Page: 64

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Curriculum Focus: Conversion factors, unit cancellations, density problems [d = m/V], and multi-step metric unit conversions.

- Physical Quantities & SI System: SI Base Units in Chemistry: Mass [kilogram, kg], Length [meter, m], Time [second, s], Temperature [Kelvin, K = C + 273.15], Amount of substance [mole, mol = 6.022 x 10^2^3 particles]. Derived units: Volume [m^3, L = 1 dm^3 = 1000 cm^3], Density [d = m / V in g/cm^3 or kg/m^3], Concentration [mol/L].

- Scientific Notation & Metric Prefixes: Scientific notation N x 10^n where 1 <= N < 10. Metric prefixes: Mega [M, 10], Kilo [k, 10^3], Centi [c, 10^2], Milli [m, 10^3], Micro [mu, 10], Nano [n, 10].

- Significant Figures [Sig Figs] & Uncertainty: Non-zero digits are significant. Trapped zeros [101] are significant. Leading zeros [0.005] are NOT significant. Trailing zeros after decimal [2.50] ARE significant. Arithmetic rules: Addition/Subtraction matches fewest decimal places; Multiplication/Division matches fewest total sig figs.

- Dimensional Analysis [Factor-Label Method]: Converting units using conversion factor ratios equal to 1 [e.g., 1 inch = 2.54 cm]. Canceling unwanted units algebraically in sequential steps.

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Chemical measurements mandate standard units and dimensional consistency. Factor-label conversions ensure error-free calculations across metric scales from sub-nanometer atomic radii to kiloton industrial yields.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Always convert temperature to absolute Kelvin K = C + 273.15 in gas law calculations.
- 1 mL = 1 cm^3 and 1 L = 1000 cm^3 = 1 dm^3.
- In multiplication/division, the final answer cannot have more significant figures than the measurement with the fewest sig figs.

Ethiopian Context: Analytical quality control laboratories at the Ethiopian Pharmaceuticals Manufacturing SC [EPHARM] employ dimensional analysis for drug dosage concentrations.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 65

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 66

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 67

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 68

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 69

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 70

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 71

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 72

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 73

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.1 Historical Development of the Atomic Model

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 66 - Current Page: 74

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Curriculum Focus: Dalton atomic theory, Thomson cathode ray experiment and plum pudding model, Rutherford gold foil experiment and nuclear atom.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.2 Subatomic Particles: Protons, Neutrons and Electrons

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 75 - Current Page: 75

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Curriculum Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.2 Subatomic Particles: Protons, Neutrons and Electrons

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 75 - Current Page: 76

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Curriculum Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.2 Subatomic Particles: Protons, Neutrons and Electrons

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 75 - Current Page: 77

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Curriculum Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.2 Subatomic Particles: Protons, Neutrons and Electrons

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 75 - Current Page: 78

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Curriculum Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.2 Subatomic Particles: Protons, Neutrons and Electrons

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 75 - Current Page: 79

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Curriculum Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.2 Subatomic Particles: Protons, Neutrons and Electrons

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 75 - Current Page: 80

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Curriculum Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.2 Subatomic Particles: Protons, Neutrons and Electrons

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 75 - Current Page: 81

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Curriculum Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.2 Subatomic Particles: Protons, Neutrons and Electrons

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 75 - Current Page: 82

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Curriculum Focus: Charges, relative masses, and locations of particles within the atomic nucleus and extranuclear cloud.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.3 Atomic Number, Mass Number, and Isotopes

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 83 - Current Page: 83

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Curriculum Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.3 Atomic Number, Mass Number, and Isotopes

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 83 - Current Page: 84

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Curriculum Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.3 Atomic Number, Mass Number, and Isotopes

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 83 - Current Page: 85

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Curriculum Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.3 Atomic Number, Mass Number, and Isotopes

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 83 - Current Page: 86

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Curriculum Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.3 Atomic Number, Mass Number, and Isotopes

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 83 - Current Page: 87

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Curriculum Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.3 Atomic Number, Mass Number, and Isotopes

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 83 - Current Page: 88

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Curriculum Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.3 Atomic Number, Mass Number, and Isotopes

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 83 - Current Page: 89

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Curriculum Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.3 Atomic Number, Mass Number, and Isotopes

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 83 - Current Page: 90

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Curriculum Focus: Z and A notation, calculation of neutrons [A - Z], isotopes of hydrogen, carbon, and chlorine, and calculating relative atomic mass.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.4 Bohr Model and Electron Energy Levels

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 91 - Current Page: 91

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Curriculum Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.4 Bohr Model and Electron Energy Levels

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 91 - Current Page: 92

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Curriculum Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.4 Bohr Model and Electron Energy Levels

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 91 - Current Page: 93

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Curriculum Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.4 Bohr Model and Electron Energy Levels

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 91 - Current Page: 94

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Curriculum Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.4 Bohr Model and Electron Energy Levels

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 91 - Current Page: 95

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Curriculum Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.4 Bohr Model and Electron Energy Levels

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 91 - Current Page: 96

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Curriculum Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.4 Bohr Model and Electron Energy Levels

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 91 - Current Page: 97

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Curriculum Focus: Quantized energy shells [K, L, M, N], electron configuration [2n^2 rule] and sublevels [s, p, d, f] for elements Z = 1 to 20.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.5 Valence Electrons and Electronic Stability

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 98 - Current Page: 98

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Curriculum Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.5 Valence Electrons and Electronic Stability

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 98 - Current Page: 99

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Curriculum Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.5 Valence Electrons and Electronic Stability

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 98 - Current Page: 100

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Curriculum Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.5 Valence Electrons and Electronic Stability

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 98 - Current Page: 101

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Curriculum Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.5 Valence Electrons and Electronic Stability

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 98 - Current Page: 102

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Curriculum Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.5 Valence Electrons and Electronic Stability

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 98 - Current Page: 103

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Curriculum Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

3.5 Valence Electrons and Electronic Stability

Unit 3 [Pages 65-104] - Topic Starts on Textbook Page 98 - Current Page: 104

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Curriculum Focus: Outermost shell electrons, Lewis dot symbols, and relationship between electron configuration and chemical reactivity.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.1 Historical Development and the Modern Periodic Law

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 106 - Current Page: 105

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Curriculum Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.1 Historical Development and the Modern Periodic Law

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 106 - Current Page: 106

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Curriculum Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.1 Historical Development and the Modern Periodic Law

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 106 - Current Page: 107

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Curriculum Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.1 Historical Development and the Modern Periodic Law

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 106 - Current Page: 108

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Curriculum Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.1 Historical Development and the Modern Periodic Law

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 106 - Current Page: 109

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Curriculum Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.1 Historical Development and the Modern Periodic Law

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 106 - Current Page: 110

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Curriculum Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.1 Historical Development and the Modern Periodic Law

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 106 - Current Page: 111

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Curriculum Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.1 Historical Development and the Modern Periodic Law

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 106 - Current Page: 112

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Curriculum Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.1 Historical Development and the Modern Periodic Law

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 106 - Current Page: 113

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Curriculum Focus: Dobereiner triads, Newlands octaves, Mendeleev periodic table vs Moseley modern periodic law based on atomic number.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.2 Periods, Groups, and Blocks of the Periodic Table

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 114 - Current Page: 114

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Curriculum Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.2 Periods, Groups, and Blocks of the Periodic Table

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 114 - Current Page: 115

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Curriculum Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.2 Periods, Groups, and Blocks of the Periodic Table

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 114 - Current Page: 116

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Curriculum Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.2 Periods, Groups, and Blocks of the Periodic Table

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 114 - Current Page: 117

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Curriculum Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.2 Periods, Groups, and Blocks of the Periodic Table

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 114 - Current Page: 118

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Curriculum Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.2 Periods, Groups, and Blocks of the Periodic Table

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 114 - Current Page: 119

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Curriculum Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.2 Periods, Groups, and Blocks of the Periodic Table

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 114 - Current Page: 120

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Curriculum Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.2 Periods, Groups, and Blocks of the Periodic Table

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 114 - Current Page: 121

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Curriculum Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.2 Periods, Groups, and Blocks of the Periodic Table

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 114 - Current Page: 122

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Curriculum Focus: Periods [1-7], Groups [IA-VIIIA and transition elements], s-, p-, d-, and f-blocks; metals, nonmetals, and metalloids.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.3 Periodic Trends: Atomic Radius and Ionic Size

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 123 - Current Page: 123

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Curriculum Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.3 Periodic Trends: Atomic Radius and Ionic Size

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 123 - Current Page: 124

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Curriculum Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.3 Periodic Trends: Atomic Radius and Ionic Size

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 123 - Current Page: 125

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Curriculum Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.3 Periodic Trends: Atomic Radius and Ionic Size

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 123 - Current Page: 126

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Curriculum Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.3 Periodic Trends: Atomic Radius and Ionic Size

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 123 - Current Page: 127

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Curriculum Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.3 Periodic Trends: Atomic Radius and Ionic Size

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 123 - Current Page: 128

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Curriculum Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.3 Periodic Trends: Atomic Radius and Ionic Size

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 123 - Current Page: 129

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Curriculum Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.3 Periodic Trends: Atomic Radius and Ionic Size

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 123 - Current Page: 130

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Curriculum Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.3 Periodic Trends: Atomic Radius and Ionic Size

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 123 - Current Page: 131

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Curriculum Focus: Effective nuclear charge [Z_eff], electron shielding, atomic radius decrease across a period and increase down a group.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 132

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 133

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 134

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 135

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 136

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 137

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 138

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 139

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 140

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 141

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

4.4 Ionization Energy, Electron Affinity, and Electronegativity

Unit 4 [Pages 105-142] - Topic Starts on Textbook Page 132 - Current Page: 142

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Curriculum Focus: First ionization energy, electron affinity, Pauling electronegativity scale, and metallic vs nonmetallic character trends.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.1 The Octet Rule and Why Atoms Combine

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 144 - Current Page: 143

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Curriculum Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.1 The Octet Rule and Why Atoms Combine

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 144 - Current Page: 144

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Curriculum Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.1 The Octet Rule and Why Atoms Combine

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 144 - Current Page: 145

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Curriculum Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.1 The Octet Rule and Why Atoms Combine

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 144 - Current Page: 146

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Curriculum Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.1 The Octet Rule and Why Atoms Combine

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 144 - Current Page: 147

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Curriculum Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.1 The Octet Rule and Why Atoms Combine

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 144 - Current Page: 148

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Curriculum Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.1 The Octet Rule and Why Atoms Combine

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 144 - Current Page: 149

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Curriculum Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.1 The Octet Rule and Why Atoms Combine

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 144 - Current Page: 150

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Curriculum Focus: Noble gas configuration [ns^2np], energy lowering during bond formation, and Lewis electron dot representations.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.2 Ionic [Electrovalent] Bonding and Properties

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 151 - Current Page: 151

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Curriculum Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.2 Ionic [Electrovalent] Bonding and Properties

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 151 - Current Page: 152

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Curriculum Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.2 Ionic [Electrovalent] Bonding and Properties

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 151 - Current Page: 153

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Curriculum Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.2 Ionic [Electrovalent] Bonding and Properties

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 151 - Current Page: 154

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Curriculum Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.2 Ionic [Electrovalent] Bonding and Properties

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 151 - Current Page: 155

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Curriculum Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.2 Ionic [Electrovalent] Bonding and Properties

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 151 - Current Page: 156

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Curriculum Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.2 Ionic [Electrovalent] Bonding and Properties

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 151 - Current Page: 157

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Curriculum Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.2 Ionic [Electrovalent] Bonding and Properties

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 151 - Current Page: 158

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Curriculum Focus: Electron transfer from metal to nonmetal, crystal lattice energy, high melting points, and electrical conductivity of molten/aqueous ions.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.3 Covalent Bonding and Molecular Geometry

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 159 - Current Page: 159

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Curriculum Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.3 Covalent Bonding and Molecular Geometry

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 159 - Current Page: 160

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Curriculum Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.3 Covalent Bonding and Molecular Geometry

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 159 - Current Page: 161

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Curriculum Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.3 Covalent Bonding and Molecular Geometry

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 159 - Current Page: 162

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Curriculum Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.3 Covalent Bonding and Molecular Geometry

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 159 - Current Page: 163

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Curriculum Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.3 Covalent Bonding and Molecular Geometry

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 159 - Current Page: 164

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Curriculum Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.3 Covalent Bonding and Molecular Geometry

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 159 - Current Page: 165

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Curriculum Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.3 Covalent Bonding and Molecular Geometry

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 159 - Current Page: 166

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Curriculum Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.3 Covalent Bonding and Molecular Geometry

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 159 - Current Page: 167

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Curriculum Focus: Sharing electron pairs between nonmetals; single, double, triple, and coordinate [dative] covalent bonds.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 168

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 169

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 170

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 171

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 172

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 173

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 174

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 175

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 176

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 177

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 178

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 179

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 180

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.

5.4 Polar Bonds, Metallic Bonding, and Intermolecular Forces

Unit 5 [Pages 143-181] - Topic Starts on Textbook Page 168 - Current Page: 181

1. CORE CURRICULUM DEFINITIONS & PRINCIPLES

Topic Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Curriculum Focus: Bond polarity, dipole-dipole forces, London dispersion forces, hydrogen bonding in H_2O/NH_3/HF, and the electron-sea model of metals.

- Evolution of Atomic Models: Dalton [indivisible solid sphere model] Thomson [Cathode ray tube experiment, discovered electron e, "Plum Pudding" model] Rutherford [Gold foil alpha-particle scattering experiment, discovered dense positive nucleus] Bohr [Quantized circular electron orbits / energy levels K, L, M, N] Quantum Mechanical Model [Electron cloud orbitals].

- Subatomic Particles: Proton [p, charge +1, mass ~= 1 u, inside nucleus]; Neutron [n, charge 0, mass ~= 1 u, inside nucleus]; Electron [e, charge -1, mass ~= 1/1836 u, extranuclear space].

- Atomic Number, Mass Number & Isotopes: Atomic Number Z = number of protons [defines element identity]. Mass Number A = Protons + Neutrons [A = Z + N]. Isotopes are atoms of the SAME element with equal proton count Z but DIFFERENT neutron count N [e.g., H Protium, ^2H Deuterium, ^3H Tritium]. Relative Atomic Mass A_r = [Isotope Abundance % x Isotope Mass].

- Bohr Model & Electron Configuration: Electrons occupy discrete energy shells [n = 1, 2, 3, 4]. Maximum electron capacity per shell = 2n^2 [K=2, L=8, M=18, N=32]. Shell filling for first 20 elements [e.g., Ca Z=20: 2, 8, 8, 2].

- Structure of Modern Periodic Table: Mendeleev [ordered by atomic mass, predicted missing elements] vs Moseley Modern Periodic Law [ordered by ATOMIC NUMBER Z]. Periods [horizontal rows 1-7 = number of electron shells]; Groups [vertical columns 1-18 = valence electron count & chemical family properties].

2. ANALYTICAL EXPLANATION & SYSTEMATIC BREAKDOWN

Atomic theory reveals subatomic structure. Protons define element identity Z, neutrons dictate isotopic mass A, and extranuclear valence electrons govern chemical bonding and periodic placement.

3. KEY TAKEAWAYS & ETHIOPIAN REAL-WORLD APPLICATION

- Rutherford’s gold foil experiment proved the atom is mostly empty space with a dense, positively charged nucleus.
- Isotopes share identical chemical properties [same electron configuration] but differ in physical mass.
- Period number equals the number of occupied electron shells; Group number dictates valence electron count.

Ethiopian Context: Potash and tantalum mineral identification by the Geological Survey of Ethiopia uses relative atomic mass spectrometry.